OPENKEYS R0.5 – ONE USB-C + NFC / BLUE 16x2 LCD
Device PIN mandatory / TEN DECIMAL KEYS / NO FINGERPRINT
PCB candidate; qualification required before secret provisioning

01-usb

File: 01-usb.kicad_sch

02-power

File: 02-power.kicad_sch

03-mcu

File: 03-mcu.kicad_sch

04-se-storage

File: 04-se-storage.kicad_sch

05-nfc

File: 05-nfc.kicad_sch

06-trusted-ui

File: 06-trusted-ui.kicad_sch

07-fault-tamper

File: 07-fault-tamper.kicad_sch

08-input-guard

File: 08-input-guard.kicad_sch

09-display-power

File: 09-display-power.kicad_sch

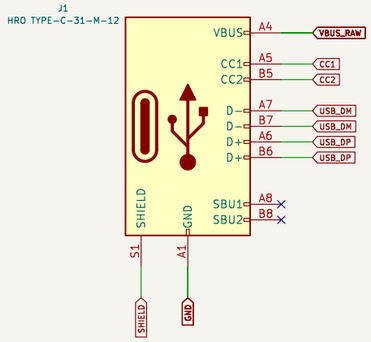
Jay-Void-BA

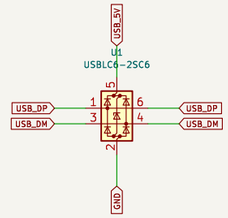
Sheet: /
File: OpenKeys.kicad_sch

Title: OpenKeys R0.5 – Engineering review

Size: A3	Date: 2026-11-10	**Rev: 0.5**
KiCad E.D.A. 9.0.2		Id: 1/10

One USB-C receptacle — USB 2.0 FS data and 5 V charging

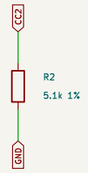


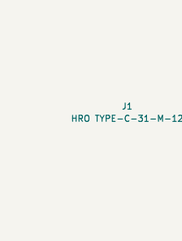


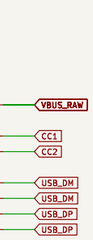


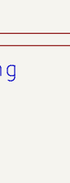


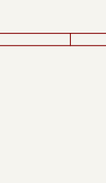




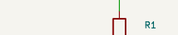


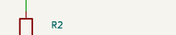












5.1k Rd on each CC pin: sink only. No PD negotiation, no 9 V request.
Apple 20 W adapter uses 5 V through a standards-compliant USB-C cable.
100mA hardware input limit: 5 V x 0.1 A = 0.50 W nominal input budget.
Input guard U14 is independent of MCU: see sheet08. Analog timing/thermal protection not yet qualified.

ENGINEERING BOARD CANDIDATE — qualify before use

Jay-Void-BA

Sheet: /01-usb/
File: 01-usb.kicad_sch

Title: OpenKeys R0.5 — 01-usb

Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 2/10

Low-rate charging, protected 500 mAh pack and 3.3 V supply

#FLG3
PWR_FLAG

VBUS_FUSED

VBUS_RAW

F1
250mA hold – qualify I2t

VBUS_FUSED

U2
BQ25185DLHR

SYS	1	SYS	
BAT	2	BAT	
CHG_STAT2_N	3	STAT2	
CHARGE_DISABLE	4	CE_N	
GND	5	GND	
BAT_NTC	6	TS_MR	
	7	ILIM_VSET	CHG_ILIM_VSET
	8	ISET	CHG_ISET
	9	STAT1	CHG_STAT1_N
	10	IN	USB_5V
	11	EP	GND

J2
Adafruit 1578 protected 1S 500mAh

BAT	1
GND	2

CHG_ILIM_VSET

R6
24k 1%

GND

CHG_ISET

R7
3.00k 1%

GND

SYS

R8
100k

CHARGE_DISABLE

CHARGE_DISABLE

CHARGE_ENABLE 1

Q2
2N7002

2

3

GND

CHARGE_ENABLE

R9
100k

GND

J6
103AT-2 external battery thermistor

BAT_NTC	1
GND	2

SW_L1

L1
1.5uH Isat>=1.5A

SW_L2

SYS

R10
0R

REG_VINA

#FLG4
PWR_FLAG

REG_VINA

U3
TPS63031DSK (3.3 V)

SYS	5	VIN	
REG_VINA	8	VINA	
REG_EN	6	EN	
GND	7	PS/SYNC	
SW_L1	4	L1	
SW_L2	2	L2	
GND	9	GND	
GND	3	PGND	
	1	VOUT	+3V3
	10	FB	+3V3

USB_5V

C2
1uF 10V

GND

BAT

C3
10uF 6.3V

GND

SYS

C4
10uF 10V

GND

REG_EN

SW1
OFF / ON

SYS

GND

SYS

C5
10uF 10V

GND

REG_VINA

C6
100nF

GND

+3V3

C7
10uF 6.3V

GND

+3V3

C8
10uF 6.3V

GND

Fixed100mA input; charge target100mA=0.2C for500mAh.
Input DPM gives system load priority, so cell charge can be slower.
CE defaults disabled. Public charging service never unlocks secrets.
Suspend: stop charge/backlight; qualify total VBUS current <=2.5mA.
Real pack PCM + attached thermistor required. No fast charge.

ENGINEERING BOARD CANDIDATE – qualify before use

Jay-Void-BA

Sheet: /02-power/
File: 02-power.kicad_sch

Title: OpenKeys R0.5 – 02-power

Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 3/10

STM32H573RIT6 — STiRoT and TrustZone protected I/O

C9
100nF

C10
100nF

C11
100nF

C12
100nF

C13
100nF

C14
4.7uF

C15
2.2uF low ESR

C16
2.2uF low ESR

C17
100nF

C18
1uF

U4
STM32H573RIT6

		VBAT	1
		VDD	19
		VDD	32
		VDD	48
		VDD	64
		VDDA	13
NRST	7	NRST	
BOOT0	60	BOOT0	
	5	PH0	
	6	PH1	
	54	PD2	
USB_GUARD_FAULT_N	8	PC0	
NRST	9	PC1	
FLASH_MISO	10	PC2	
LCD_RELEASE	11	PC3	
LCD_ARM_CLK	24	PC4	
VBUS_PRESENT_N	25	PC5	
BTN5_N	37	PC6	
BTN6_N	38	PC7	
SE_ENABLE_REQUEST	39	PC8	
LCD_HOST_SDA	40	PC9	
BTN8_N	51	PC10	
BTN9_N	52	PC11	
BTN0_N	53	PC12	
CASE_CLOSED	2	PC13	
LSE_IN	3	PC14	
LSE_OUT	4	PC15	
VCAP1	30	VCAP	
VCAP2	62	VCAP	
		PA0	14
		PA1	15
		PA2	16
		PA3	17
		PA4	20
		PA5	21
		PA6	22
		PA7	23
		PA8	41
		PA9	42
		PA10	43
		PA11	44
		PA12	45
		PA13(JTMS	46
		PA14(JTCK	49
		PA15(JTDI)	50
		PB0	26
		PB1	27
		PB2	28
		PB3(JTDO	55
		PB4(NJTRST)	56
		PB5	57
		PB6	58
		PB7	59
		PB8	61
		PB10	29
		PB12	33
		PB13	34
		PB14	35
		PB15	36
		VSS	18
		VSSA	12

CHG_STAT1_N	PA2
CHG_STAT2_N	PA3
NFC_CS_N	PA4
NFC_SCK	PA5
NFC_MISO	PA6
NFC_MOSI	PA7
LCD_HOST_SCL	PA8
USB_DM	PA11
USB_DP	PA12
SWDIO	PA13(JTMS
SWCLK	PA14(JTCK
BTN7_N	PA15(JTDI)
BTN1_N	PB0
BTN2_N	PB1
BTN3_N	PB2
SWO	PB3(JTDO
CHARGE_ENABLE	PB4(NJTRST)
BTN4_N	PB5
SE_SCL	PB6
SE_SDA	PB7
NFC_IRQ_N	PB10
FLASH_CS_N	PB12
FLASH_SCK	PB13
FLASH_MOSI	PB15

FB1
600R@100MHz

#FLG6
PWR_FLAG

R11
10k

R12
10k

C19
10nF — qualify reset timing

Y1
ABS07-32768KHZ-6-T

C20
9pF C0G — tune

C21
9pF C0G — tune

J3
Internal SWD — DNP production

+3V3	1	2	SWDIO
GND	3	4	SWCLK
GND	5	6	SWO
	7	8	
GND	9	10	NRST

STiRoT must verify protected firmware. TrustZone/GTZC must deny nonsecure access to SE, LCD, ten keys, timers, DMA and key memory.
No raw cryptography or arbitrary display commands at secure gateway.
SE_EN is a fault interlock, not proof of a correct PIN.
Separate2.2uF on VCAP1/2; no external load on VCAP.

ENGINEERING BOARD CANDIDATE — qualify before use

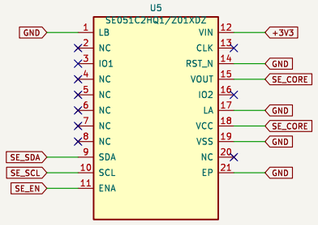
Jay-Void-BA

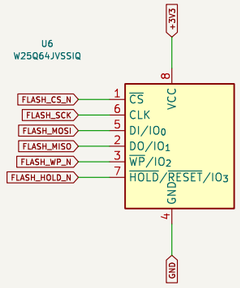
Sheet: /03-mcu/
File: 03-mcu.kicad_sch

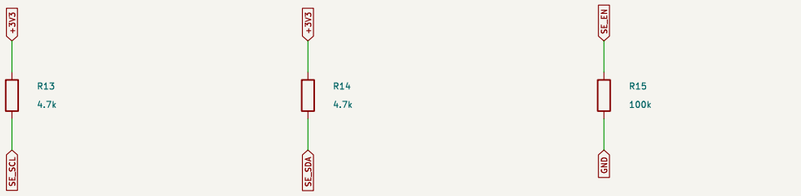
Title: OpenKeys R0.5 — 03-mcu

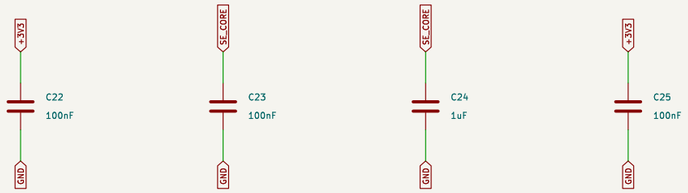
Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 4/10

Secure element, protected keys and encrypted records









Private operations: valid local six-digit PIN grant + per-request confirmation.
No biometric path. SE-native PIN ACLs and durable retries must be qualified; Root owns keypad.
Precharge attempts durably; uncertain writes deny. AEAD does not alone prevent rollback.
SE certification does not certify MCU, PCB, PIN execution path or complete product.

ENGINEERING BOARD CANDIDATE — qualify before use

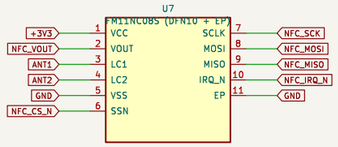
Jay-Void-BA

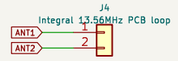
Sheet: /04-se-storage/
File: 04-se-storage.kicad_sch

Title: OpenKeys R0.5 — 04-se-storage

Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 5/10

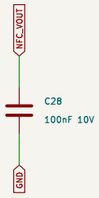
NFC Type4 card interface — battery-powered local PIN UI

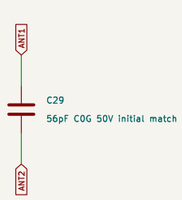
















NFC VOUT isolated from SYS/BAT/+3V3. No assumed RF-powered full device.
LCD/PIN/CTAP NFC require charged battery and device ON.
RF region: tuned coil only, no cell, metal shield or ground planes.
Mobile management needs authenticated APDUs and native OS integration.
Phone biometrics or a tap never replace device PIN verification.

ENGINEERING BOARD CANDIDATE — qualify before use

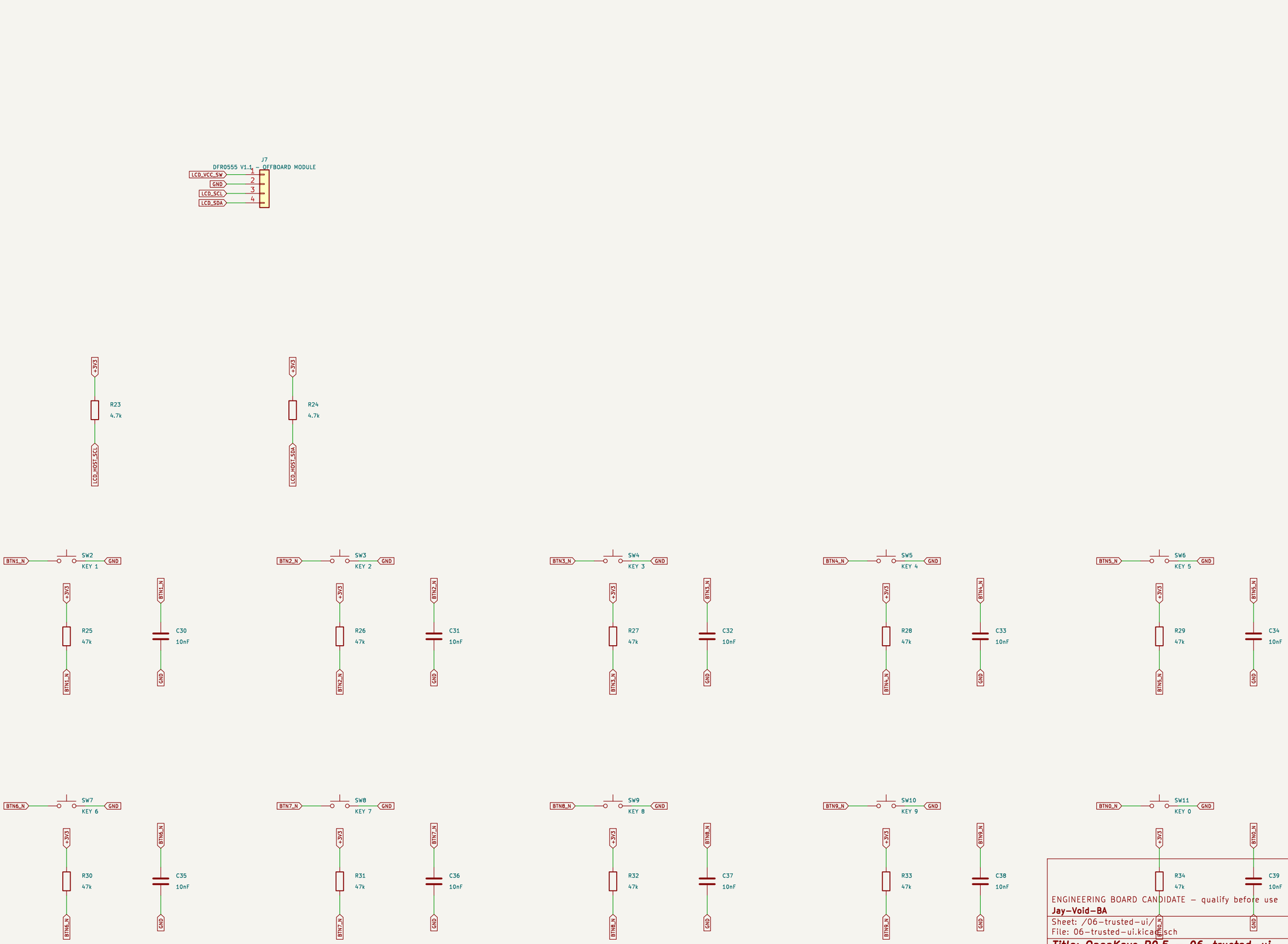
Jay-Void-BA

Sheet: /05-nfc/
File: 05-nfc.kicad_sch

Title: OpenKeys R0.5 — 05-nfc

Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 6/10

DFR0555 blue 16x2 LCD and ten independent PIN keys



ENGLISH 16x2, blue background / white text. No fingerprint parts.
PIN:6 decimal digits, masked. Hold9=erase; hold0=cancel.
Review:2=previous,8=next,5=confirm,0=cancel. New release required.
Private operations require a valid paired30s local PIN grant and exact review.
No PIN from USB/NFC/desktop. No factory/default unlock PIN.

ENGINEERING BOARD CANDIDATE – qualify before use

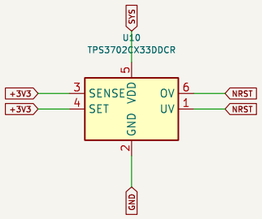
Jay-Void-BA

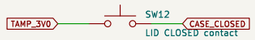
Sheet: /06-trusted-ui/
File: 06-trusted-ui.kicad_sch

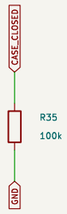
Title: *OpenKeys R0.5 – 06-trusted-ui*

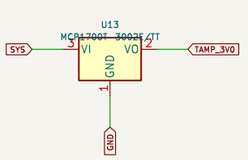
Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 7/10

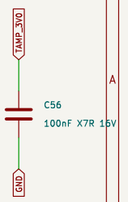
Independent reset, battery–backed case tamper and SE interlock

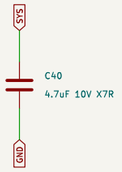


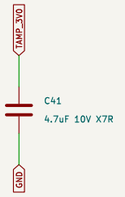


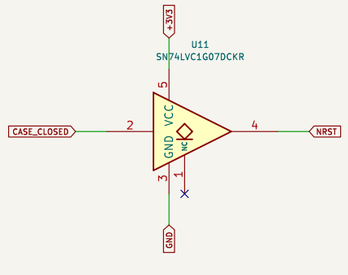


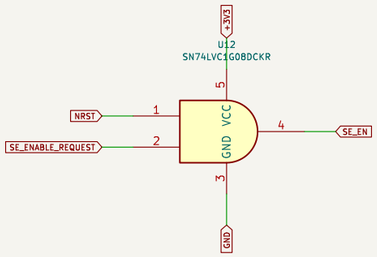




















Reset invalidates volatile grants; reboot must verify firmware and require fresh device PIN.
PC13=case input; PC1/TAMP_IN3 observes reset/fault line.
Internal BOR/PVD, clock checks and persistent precharged attempts also required.
Production debug lifecycle closed; no populated external debug connector.
VBAT+LSE must retain tamper with main OFF; full backup loss must lock, not initialize.
Measure reset/ENA latency, leakage and injection before security release.

ENGINEERING BOARD CANDIDATE – qualify before use

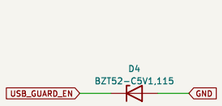
Jay–Void–BA

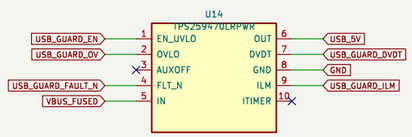
Sheet: /07–fault–tamper/
File: 07–fault–tamper.kicad_sch

Title: OpenKeys R0.5 – 07–fault–tamper

Size: A3	Date: 2026–11–10	**Rev: 0.5–board–review**
KiCad E.D.A. 9.0.2		Id: 8/10

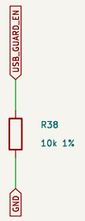
Independent VBUS overvoltage cutoff, reverse blocking and limited inrush

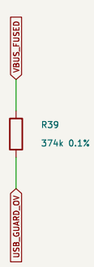


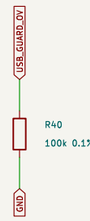


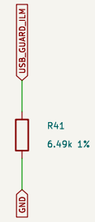


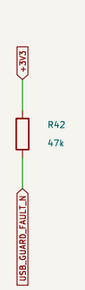


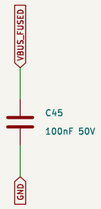


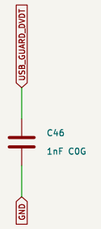












OVLO nominal1.2V x(1+374k/100k)=5.688V. Component/leakage corners in protection-calculations.json.
UVLO divider22.1k/10k: about3.85V nominal. EN is separately clamped, never raw VBUS.
6.49k ILM: about0.51A protection threshold; BQ charger still sets100mA input budget.
1nF DVDT: about2V/ms, about2.5ms ramp to5V; measure actual inrush with complete load.
F1 -> input TVS -> eFuse -> output TVS/ESD reference/charger/sense.
Independent cutoff is a circuit provision; response overshoot, CC/data faults and thermal/surge tests remain BLOCKED.

ENGINEERING BOARD CANDIDATE - qualify before use

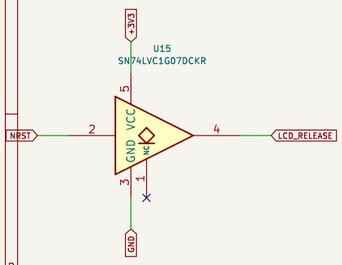
Jay-Void-BA

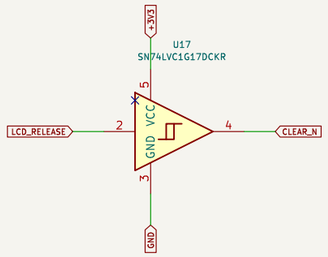
Sheet: /08-input-guard/
File: 08-input-guard.kicad_sch

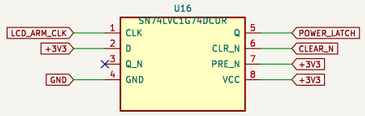
Title: OpenKeys R0.5 - 08-input-guard

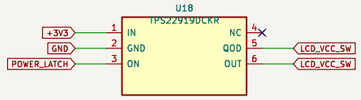
Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 9/10

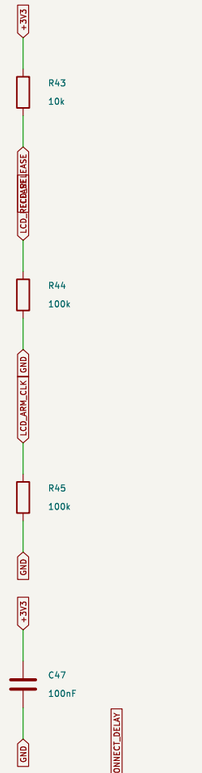
Latched LCD power removal; isolated bus; no resetless retained text

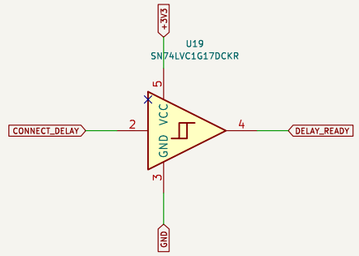


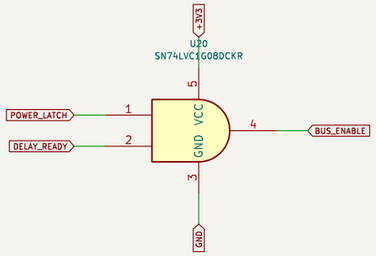


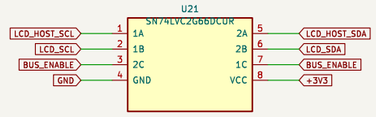


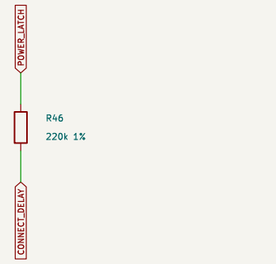


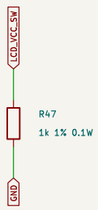


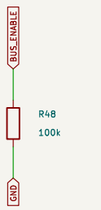


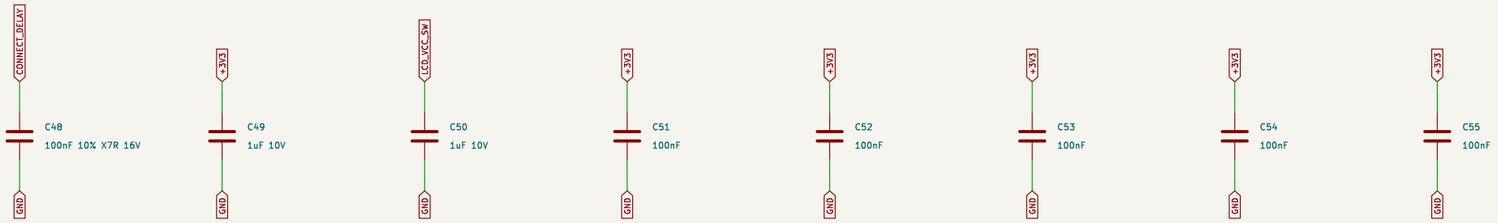












U16 D=1, PRE_N=1. NRST/PC3 low asynchronously clears Q.
Releasing reset alone does NOT restore power; a new PC4 rising edge is required.
Q controls TPS22919 ON; Q AND delayed-Q controls both analog switches.
Lock: PC4 low then PC3 low. Wait500ms before a new start; then PC3 high.
PC4 high, wait200ms before I2C. Wrong version/NAK/fault -> power OFF, latch.
3.3V nominal module supply needs load/temperature/vendor tolerance qualification.
No physical clearing, injection or production security qualification claimed.

ENGINEERING BOARD CANDIDATE — qualify before use

Jay-Void-BA

Sheet: /09-display-power/
File: 09-display-power.kicad_sch

Title: OpenKeys R0.5 — 09-display-power

Size: A3	Date: 2026-11-10	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 10/10